Supporting Information for Extra Hidden States as Kullback-Leibler Corrections under Emission Misspecification in Hidden Markov Models

Aurélien Nicosia

1 Reproducibility

The numerical study uses no external data. All random seeds are fixed in `numerics/scripts/run_review_simulation.py`, with master seed `20260610`. The code and generated outputs are available at https://github.com/AurelienNicosiaULaval/corrective-hmm-states.

The replication script compares three fitted models: a Gaussian-emission HMM with $K = 2$, a Gaussian-emission HMM with $K = 3$, and a two-state HMM with two Gaussian mixture components per state. For the replicated study, the sample sizes are $T \in \{500, 1500, 5000\}$ with 200 replications per sample size and 10 random EM starts per replicated fit. In the replicated fits, all 10 starts are screened deterministically; the three best starts are then refined with relative log-likelihood stopping tolerance 10^{-6} and maximum 100 refinement iterations. The row-level replication file is `numerics/output/review_replications.csv`; the compact summary is `numerics/output/review_summary_by_T.csv`.

2 Mixture-emission EM algorithm

For hidden state $k \in \{1, \dots, K\}$ and mixture component $m \in \{1, \dots, M\}$, the fitted emission density is

$$b_k(y) = \sum_{m=1}^{M} \rho_{km}\, \phi(y; \mu_{km}, \sigma^2_{km}), \qquad \rho_{km} \geq 0, \quad \sum_{m=1}^{M} \rho_{km} = 1.$$

The hidden chain has initial probabilities π_k and transition matrix $A = (a_{k\ell})$. Given current parameters, the scaled forward-backward recursion gives the state posterior probabilities

$$\gamma_t(k) = P(X_t = k \mid Y_{1:T})$$

and transition posterior sums

$$\xi(k, \ell) = \sum_{t=2}^{T} P(X_{t-1} = k, X_t = \ell \mid Y_{1:T}).$$

Within each state, the component responsibility is

$$\tau_t(k, m) = \frac{\gamma_t(k)\rho_{km}\phi(Y_t; \mu_{km}, \sigma^2_{km})}{\sum_{r=1}^{M} \rho_{kr}\phi(Y_t; \mu_{kr}, \sigma^2_{kr})}.$$

The M-step is

$$\pi_k^{new} = \gamma_1(k), \qquad a_{k\ell}^{new} = \frac{\xi(k, \ell)}{\sum_r \xi(k, r)},$$

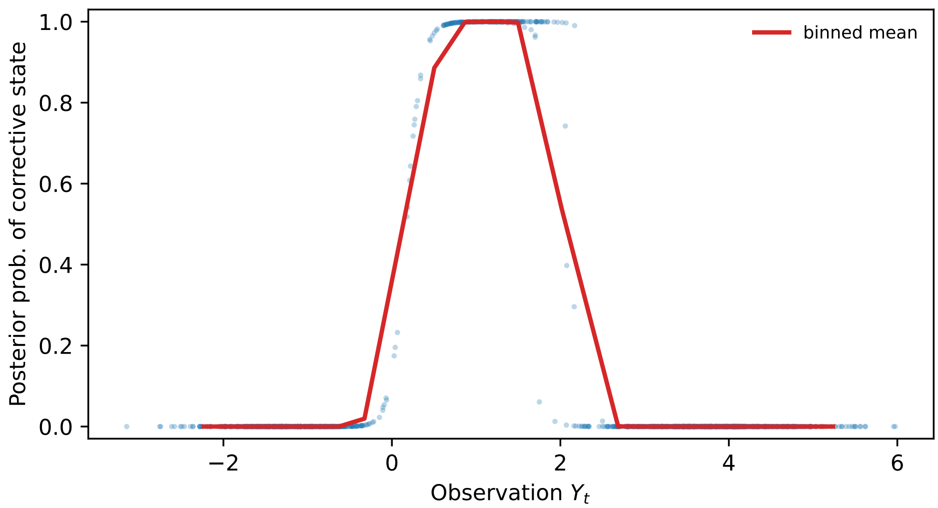

Figure 1: Posterior probability of the corrective Gaussian-HMM state in the reference $K = 3$ fit.

$$\rho_{km}^{new} = \frac{\sum_t \tau_t(k,m)}{\sum_t \gamma_t(k)}, \qquad \mu_{km}^{new} = \frac{\sum_t \tau_t(k,m)Y_t}{\sum_t \tau_t(k,m)},$$
$$(\sigma_{km}^2)^{new} = \max\left\{\sigma_{\min}^2, \frac{\sum_t \tau_t(k,m)(Y_t - \mu_{km}^{new})^2}{\sum_t \tau_t(k,m)}\right\}.$$

The implementation uses multiple random starts, scaled likelihoods, relative log-likelihood stopping tolerance 10^{-6}, a minimum standard deviation $\sigma_{\min} = 0.05$, and sorts states by their fitted state-level means after optimization. In the replicated study, the Gaussian-emission fits screen each start for 25 EM iterations, the mixture-emission fits screen each start for 30 EM iterations, and the best three starts are refined. For the model used in the paper, $K = 2$ and $M = 2$. The BIC parameter count is

$$(K-1) + K(K-1) + K\{(M-1) + 2M\},$$

corresponding to the initial probabilities, transition probabilities, mixture weights, means, and standard deviations.

3 Supplementary diagnostics

Let

$$\Delta_G = \text{BIC}(\text{Gaussian } K = 3) - \text{BIC}(\text{Gaussian } K = 2),$$

and let

$$\Delta_M = \text{BIC}(\text{Mixture } K = 2, M = 2) - \text{BIC}(\text{Gaussian } K = 3).$$

Thus $\Delta_G < 0$ favours three Gaussian states within the simple Gaussian emission family, whereas $\Delta_M < 0$ favours the enriched two-state mixture-emission model over the three-state Gaussian model. The main text gives the selection-frequency table and decoding figure; the full row-level outputs are supplied as CSV files.

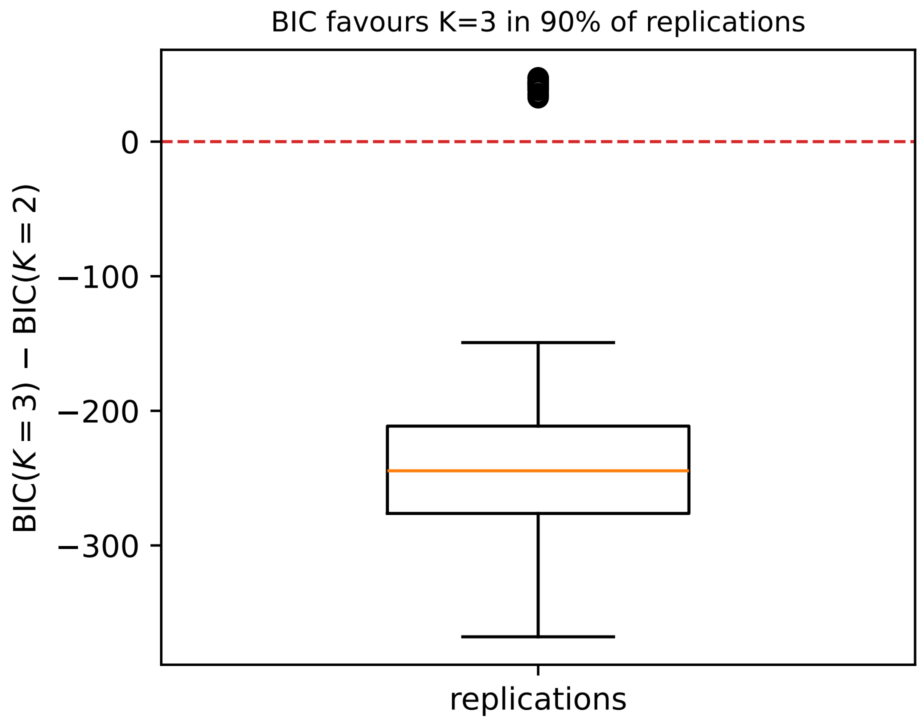

Figure 2: Distribution of Δ_G for the $T = 1500$ replicated study.

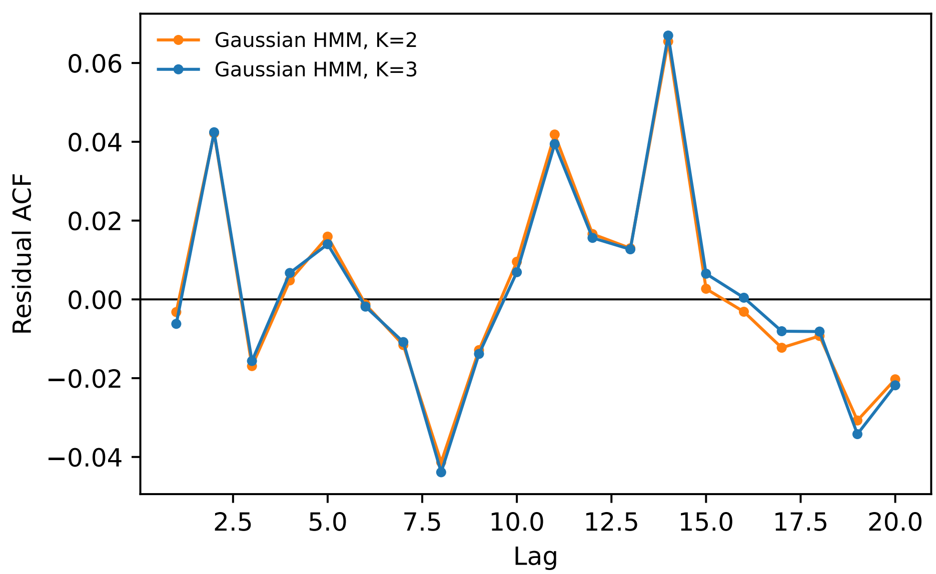

Figure 3: Autocorrelation of approximate one-step standardized prediction residuals for the Gaussian-emission HMMs with $K = 2$ and $K = 3$.